

AeroGrid Engineering Report – Wind Turbine Telemetry Analysis



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Section 1: Findings

This report will firstly examine the results of the telemetry analysis.

Table 1 presents the findings of the data which is downloaded into a succinct CSV file.

Table 1: Vibration and Temperature data with anomaly decision

Turbine ID	Average Temperature (°C)	Maximum Vibration (mm/s)	Temperature Alert	Vibration Alert
T-01	65.324	11.9	No	No
T-02	65.157	12	No	No
T-03	65.315	12	No	No
T-04	90.581	10	Yes	No
T-05	64.911	12	No	No
T-06	64.804	12	No	No
T-07	67.781	25	No	Yes
T-08	64.858	12	No	No
T-09	65.427	12	No	No

Table 1 emphasises how turbine 'T-07' should be assessed for excessive vibration and turbine 'T-04' should be assessed for excessive temperature. Figure 1 and Figure 2, show the readings compared to the acceptable metric value.

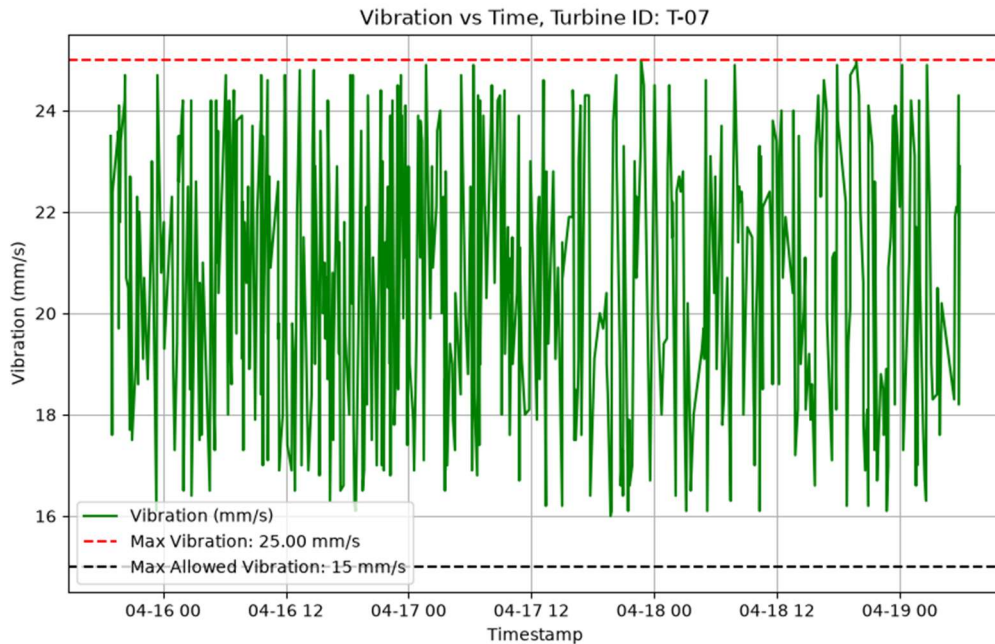


Figure 1: Vibration vs Time for Turbine T-07

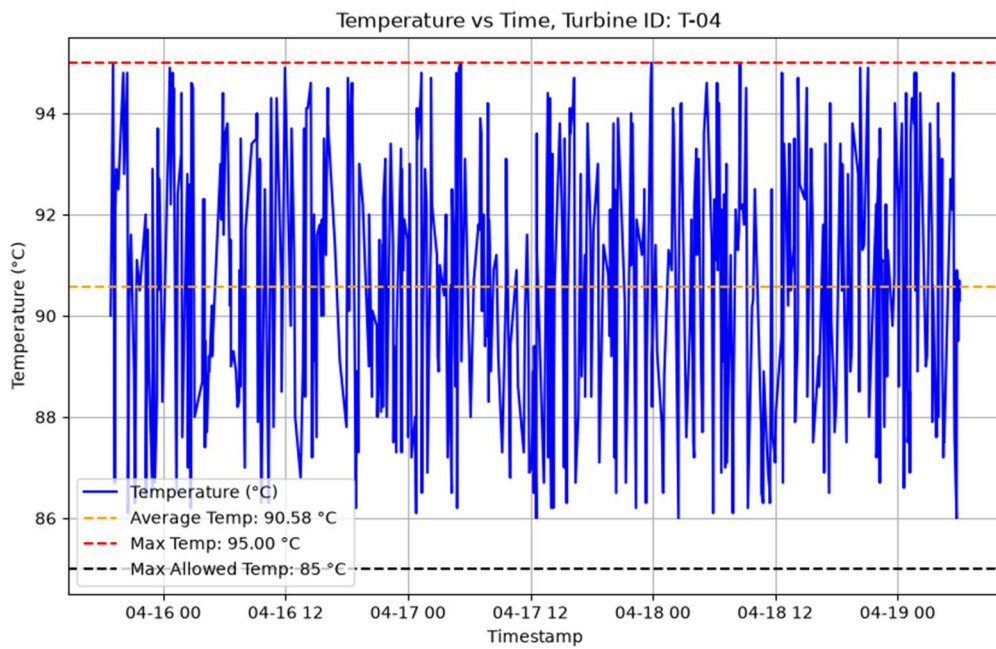


Figure 2: Temperature vs Time for Turbine T-04

Section 2: Architecture Justification

Figure 3 shows the system architecture for this analysis:

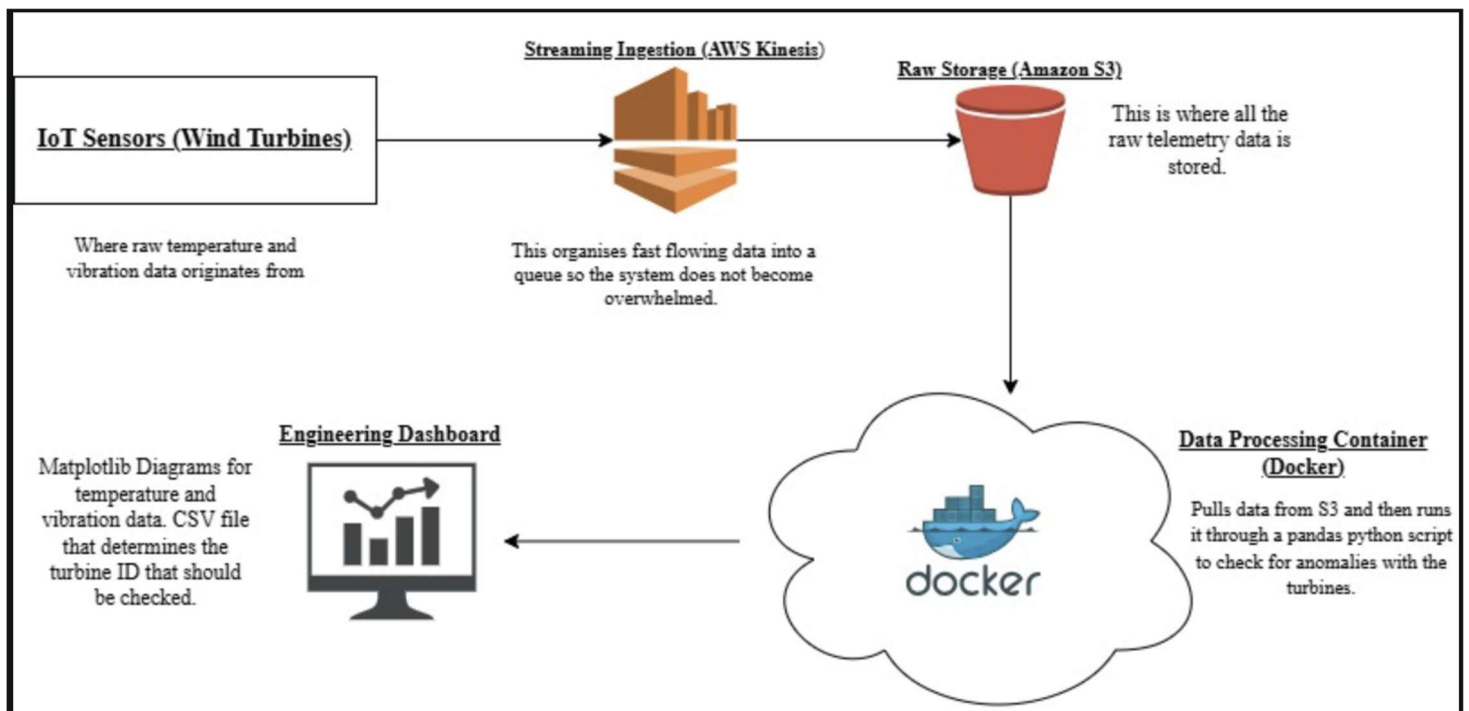


Figure 3: System Architecture Diagram

Firstly, there are the IoT sensors which are mounted on the turbines to extract the raw data (RPM, temperature and vibration) at specific millisecond timestamps. Then, the data is ingested by AWS kinesis which catches chaotic data and organises it into a queue. This prevents downstream servers from becoming overwhelmed hence solving the crashing issue.

Amazon Simple Storage Service is the highly durable cloud service system used. Once Kinesis catches the data, it is immediately stored into S3. This means that AeroGrid has permanent, historical record of all data collected.

For processing, docker and ECS are used. The python docker data is processed using a pandas script the identifies all turbines producing anomalies. This is then sent to an output engineering dashboard where diagrams of temperature and vibration data are saved along with a CSV file such as *Table 1*.

Section 3: Cost Optimisation

Firstly, by using AWS ECS and Docker, the processing engine does not have to stay on 24/7 and is only used batchwise. This means you are only billed on the times the code is actively running. Secondly, AWS S3 is incredibly cheap as bulk storage.